KK-L-164
June 23, 1964

SUPERSEDING
Fed. Spec. KK-L-157
July 31, 1956,
Fed. Spec. KK-L-163
January 2, 1957, and
Fed. Spec. KK-L-231a
November 8, 1949

FEDERAL SPECIFICATION

LEATHER, CATTLEHIDE, PACKING, VEGETABLE-TANNED, CHROME-TANNED, CHROME-VEGETABLE RETANNED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for cattlehide leather used in the preparation of packings and gaskets.

1.2 Classification.

1.2.1 *Tannage.* The packing leather shall be of the following tannages, as specified (see 6.2).

Tannage I—Vegetable-Tanned.
Tannage II—Chrome-Tanned.
Class 1—Regular.
Class 2—Non-Corrosive.
Tannage III—Chrome-Vegetable-Retanned.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

KK-L-311—Leather; Methods of Sampling and Testing.
QQ-S-781—Steel Strapping, Flat.
UU-P-268—Paper, Kraft, Untreated, Wrapping.
UU-P-271—Paper, Wrapping, Waterproofed Kraft.
VV-G-100—Gasoline, Unleaded.
CCC-C-429—Cloth, Osnaburg, Cotton.
CCC-C-467—Cloth, Jute (or Kenaf), Burlap.
PPP-B-636—Box, Fiberboard.
PPP-P-291 — Paperboard, Wrapping, Cushioning.
PPP-T-60 — Tape; Pressure-Sensitive Adhesive, Waterproofed for Packaging and Sealing.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

FSC 8330

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-T-40625—Tubing, Bias Sewn Burlap or Osnaburg Cloth.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards.

ASTM D-471—Vulcanizates Resulting from Immersion in Liquids.
ASTM D-1611—Corrosion Produced by Leather in Contact with Metal.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

3. REQUIREMENTS

3.1 Material. The packing leather shall be made from cattlehides furnished in the form of bends or belting butt bends.

3.2 Tannage.

3.2.1 *Tannage I.* The leather shall be tanned by a vegetable tanning process.

3.2.2 *Tannage II.* The leather shall be tanned by a chrome tanning process.

3.2.3 *Tannage III.* The leather shall be chrome tanned followed by a vegetable tannage.

3.3 Finish. The leather shall be full grain. The flesh side shall be free from flesh or loose fibers and may be shaved, skived or split to secure greater uniformity of thickness.

3.4 Trim (form) and area. The trim of the bends or belting butt bends shall be as specified by the definitions for "bends" and "belting butt bends" in section 5 of KK-L-311. The area expressed in terms of square feet shall be legibly marked on the butt area of the flesh side of the leather.

3.5 Color. The color of the packing leather shall be natural as produced by the tanning agents.

3.6 Physical requirements.

3.6.1 *Weight and thickness.* The leather shall be of the weight and thickness as specified (see 6.2) and as shown in table I.

TABLE I.—*Thickness*

Weight	Thickness (64th inch)
Light	4 and below 6
Medium	6 and below 10
Heavy	10 and below 13
Extra heavy	13 and below 17

3.6.2 *Breaking load.* Not less than 80 percent of the specimens tested shall have breaking load values of not less than that specified in table II when tested as specified in 4.3.

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TABLE II.—*Breaking load*

Thickness, inches	Breaking load, pounds		
Tannage	*I*	*II*	*III*
4/64 and below 6/64	65	80	60
6/64 and below 10/64	90	110	85
10/64 and below 13/64	120	140	110
13/64 and below 17/64	150	170	135

3.6.3 *Cracking.* The leather shall not crack when tested as specified in 4.3.

3.6.4 *Shrinkage temperature.*

3.6.4.1 *Tannage I.* The shrinkage temperature of the leather from tannage I shall be not less than 75°C. when tested as specified in 4.3.

3.6.4.2 *Tannage II and III.* The shrinkage temperature of the leather from tannages II and III shall be not less than 99° C. when tested as specified in 4.3.

3.6.5 *Area stability, tannages II and III.* When tested under the conditions outlined in table III, the maximum area shrinkages shall be those given in table III.

TABLE III.

Temperature	Time	Media	Maximum area shrinkage
			Percent
70°C.	1 hr.	Gasoline	2
100°C.	2 min.	Oil	10
95°C.	1 hr.	Air	5
Boiling water	5 min.	Water	5

3.7 **Chemical requirements.**

3.7.1 *Fungicide treatment.* When specified, (see 6.2), the leather shall be treated with the fungicide paranitrophenol. The leather shall contain 0.18 to 0.70 percent paranitrophenol based upon the dry weight of the leather when tested as specified in 4.3.

3.7.2 *Tannage II, class* 2 (*see* 6.2). When tested in accordance with ASTM method D-1611, the chrome-tanned leather shall be non-corrosive.

3.7.3 The leather shall conform to the chemical requirements specified in table IV when tested as specified in 4.3.

3.8 **Selection.** The leather shall be free from deep cuts, damaged grain, open scratches, grub holes, brands, hip marks and rough shoulders, or any other defects which affect cutting value greater than 5

TABLE IV.—*Chemical requirements*

Characteristics	Tannage I		Tannage II				Tannage III	
			Class 1		Class 2			
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Total ash (percent)[1]	—	2.0	—	—	—	—	—	—
Chloroform extract (percent)[1]	12.0	20.0	3.0	10.0	3.0	10.0	7.0	15.0
Water soluble materials (percent)[1]	—	12.0	—	—	—	—	—	—
Degree of tannage[1]	—	65.0	—	—	—	—	—	—
Acidity (pH)	3.3	—	3.3	—	3.3	—	3.3	—
Chromic oxide (percent)[2]	—	—	4.5	—	4.5	—	3.3	—
Total ash, percent of chromic oxide[1]	—	—	—	130	—	120	—	130

[1] On moisture-free basis.
[2] On moisture- and grease-free basis.

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percent of the total area of the bends or belting butt bends.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for meeting all requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections or tests set forth in the specifications where such inspections or tests are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection. Sampling for inspection shall be in accordance with MIL-STD-105 except as otherwise indicated herein.

4.2.1 *Inspection of the end item.* The classification of defects found during the inspection shall be in accordance with the list shown in 4.2.1.1 and 4.2.1.2. The sample unit for these inspections shall be one bend or belting butt bend. The lot size shall be expressed in terms of one bend or belting butt bend. The inspection level and acceptable quality level for these examinations shall be in accordance with 4.2.1.3.

4.2.1.1 *Examination for visual characteristics.*

Examine	Defect
Material	Not as specified.
Finish	Not full grain, flesh side not smooth or contains areas of loose flesh or coarse fiber.
Trim	Not bends or belting butt bends as specified.
Color	Not natural color as derived from tanning agents.
Selection	Deep cuts, open scratches, damaged grain, open grub holes, brands, hip marks, rough shoulders or other defects impairing cutting value of more than 5 percent of the area of the bend or belting butt bend.

Note: Open grain, minor cuts, healed briar marks, soft scars, light well-healed scratches or grubs, light fat wrinkles or slight salt stains shall not be classified as defects.

4.2.1.2 *Examination for dimensional characteristics.*

Examine	Defect
Thickness	Thickness of leather not as specified.

Note: Method 1011 KK-L-311 shall be used for thickness except that the presser foot of the spring gage shall be snapped to contact the surface of the leather.

Area (expressed in terms of square feet)	Not as marked on the bend or belting butt bend.

4.2.1.3 *Inspection levels and acceptable quality levels (AQL's).* The inspection levels and acceptable quality levels, expressed in defects per hundred units shall be as follows:

Examination	Inspection level	AQL's
For defects applicable to 4.2.1.1, visual characteristics	II	4.0
For defects applicable to 4.2.1.2, dimensional characteristics	L-6	4.0

4.2.2 *Examination of preparation for delivery requirements.* An examination shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end inspection lot. The inspection level shall be L-4 and the AQL shall be 4.0 defects per 100 units.

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Examine	Defect
Marking (exterior and interior) ...	Omitted, incorrect, illegible, or improper size, location, sequence, or method of application.
Materials ...	Any component missing.
	Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete wrapping and taping or tying of rolls, incomplete closure of container flaps, loose strapping, inadequate stapling.
	Bulging or distortion of containers. Closure of cloth covered bundles incorrect, affecting serviceability.
Content	Not rolled grain out in sets of 3 or 6 and the final piece of each roll not rolled flesh side out.
Weight	Net weight exceeds requirement.

4.3 **Sampling for testing.** Sampling procedures and the location from which the sample unit is to be obtained shall be in accordance with section 6 of KK-L-311. The specimen for testing strength requirements shall be selected from the corner of the test sample unit which is closest to the first "A" in the word "Area" of figure 6, section 6 of KK-L-311. The methods of testing specified in this Specification wherever applicable and as listed in table V shall be followed. When chemical requirements are specified on a moisture-free basis or a hide substance basis, the method for moisture determination shall be method 6221 of KK-L-311 and the method for hide substance determination shall be method 6461 of KK-L-311. The lot shall be rejected if any one of the following conditions exist:

a. More than three test failures occur for either the breaking load or area stability requirements.

b. More than one test failure occurs for the remaining requirements applicable to the sample unit.

c. Any average value from a sample representing the lot which fails to meet the specified requirement.

TABLE V.—*Tests*

Characteristic	Requirement paragraph	Test method (Spec. KK-L-311)	Requirement applicable to	
			Sample unit	Sample average
Material[1]	3.1	Std. Comm'l	X	—
Tannage[1]	3.2	Std. Comm'l	X	—
Breaking load	3.6.2	4.3.1	X	—
Cracking	3.6.3	4.3.2	X	—
Shrinkage temperature	3.6.4	7011[2]	—	X
Area stability	3.6.5	4.3.3	X	—
Paranitrophenol[4]	3.7.1	4.3.4	—	X
Corrosion	3.7.2	ASTM D-1611	X	—
Total ash[4]	3.7.3	6421[3]	—	X
Chloroform extract[4]	3.7.3	6311	—	X
Water-soluble materials[4]	3.7.3	6341	—	X
Degree of tannage[4]	3.7.3	6641	—	X
Acidity (pH)	3.7.3	6621	—	X
Chromic oxide[5]	3.7.3	6515	—	X

[1] The supplier shall submit a certificate of compliance for this characteristic.

[2] Except that the specimen shall be immersed in the heating medium for not less than two hours, plus or minus fifteen minutes, prior to being tested and that the heating medium shall be water.

[3] Total ash in tannage II and III shall be expressed in percent of the chromic oxide present in the leather.

[4] On moisture-free basis.

[5] On a moisture- and grease-free basis.

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4.3.1 *Breaking load.* Method 2011 of KK-L-311 shall be used except that only the load at break need be determined. The cross sectional area of the specimen and the calculation of tensile strength need not be determined. The breaking load of the specimen shall be recorded to the nearest pound.

4.3.2 *Cracking.* Method 4021 of KK-L-311 shall be used except that the results shall be reported as pass or fail. The diameter of the mandrel used shall be as follows:

Weight of leather	Diameter of mandrel
Light	1/4 inch
Medium	3/8 inch
Heavy	1/2 inch
Extra heavy	5/8 inch

4.3.3 *Area stability.*

4.3.3.1 *Immersion media.*

a. White gasoline (unleaded) 74 octane grade, conforming to VV-G-109.

b. ASTM oil No. 1 as described in the method of test for change in vulcanizates resulting from immersion in liquids (ASTM Designation: D 471).

c. Distilled water.

d. Air.

4.3.3.2 *Test specimen.* The leather specimen shall be a piece 2 ± .03 inch by 2 ± .03 inch (5.08 ± .08 cm. by 5.08 ± .08 cm.). A small hole not in excess 0.125 inch (0.32 cm.) in diameter shall be punched on the centerline of the specimen, 0.25 inch (0.64 cm.) from the edge.

4.3.3.3 *Procedure.*

a. Measure the length of the specimen on the grain side along each of the four sides at a distance of 0.3 ± 0.1 cm. from the edge and report the average value. Compute the area by squaring the average value for the sides. Place the specimen to be measured, grain side up, on a smooth surface of contrasting color, and place a ruler divided into 0.05 cm. intervals onto the grain side, applying just enough pressure to flatten the specimen.

b. Suspend the specimens, not more than six per single test, on a copper wire in 800 ± 10 ml. of the liquid medium for a period of 16 to 24 hours at room temperature. The specimens shall be not closer than 2.0 cm. to the surface of the beaker to which the heat is applied. Space the specimens with brass nuts of 0.4 to 0.5 cn. thickness. Also prevent the specimens from touching the wall of the beaker by means of spacers.

c. Heat the fluid at the rate of 3° to 5°C. (5.5° to 9°F.) per minute until it reaches the test temperature and keep it at the test temperature for the length of time suggested in table III (measurements in air may be made by placing the specimen immediately in contact with air at the maximum temperature suggested). Use a power stirrer for oil. If water is used stir with a glass rod until the boiling temperature is reached. Care must be taken in the test with gasoline. It should be made in a hood in which fumes are rapidly carried away and a Glas-Col electric flask heater or a steam bath should be used for heating.

d. At the end of the test time remove the specimens from the fluid, wipe free of excess fluid, and allow to cool to room temperature. Compute the area as described in "a" of this paragraph and record.

e. Calculation:

Calculate the percentage area change as follows:

$$\text{Area change, per cent} = \frac{B - A}{A} \times 100$$

Where:

A = initial area, and
B = final area

4.3.4 *Test for the determination of para-nitrophenol.* Method 6711 of KK-L-311 shall be used except that 2 ml. of either chloroform extract shall be transferred to the ninth flask containing sodium citrate (20 ml. of a 10 percent solution of $Na_3C_6H_5O_7$

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in water). If the concentration of the developed paranitrophenol color is too intense for instrument readings, the 2 ml. of aliquot shall be suitably reduced. To compute the percent paranitrophenol on a dry basis use the following formula:

$$\text{Percent paranitrophenol (dry basis)} = \frac{\text{Milligrams paranitrophenol in aliquot} \times \text{dilution}}{10 \times \text{dry weight of specimen in grams}}$$

5. PREPARATION FOR DELIVERY

Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing protection for civil agency procurements.

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* Leather units, consisting of butts or butt bends, of one tannage and weight only, shall be neatly and tightly rolled together in sets of three or six. When a roll of leather consists of more than one set, each additional set shall be alternated so that the heads of one set are adjacent to the butts of the next set. The leather shall be rolled grain side out, except that the outer piece shall be rolled flesh side out. Each roll of leather shall be completely wrapped in 80-pound minimum basis weight kraft paper conforming to grade B of UU-P-268. The paper shall be folded to inclose each end of the roll and shall be secured with gummed kraft paper tape. Alternatively, each roll of leather will be completely wrapped in single-faced corrugated paperboard conforming to type II of PPP-P-291. The wrap shall be folded over to completely inclose each end of the roll. The wrap shall be secured in place by tying with jute or cotton cord or twine, applied twice around the body, approximately 12 inches from each end (when practical) and once lengthwise, tying with non-slip knots.

5.1.2 *Level C.* Leather shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Each roll of leather, packaged as specified in 5.1 shall be completely inclosed in waterproof kraft paper conforming to class B-2 of UU-P-271. The paper inclosing each end shall be tightly folded in place so the roll can be inserted into cloth covering material. The waterproofed roll of leather shall be baled in bias-sewn tubing or a bag presewn one end made from cotton osnaburg cloth conforming to type I, class 2, finish A of CCC-C-429 or jute (or kenaf) burlap cloth conforming to class 1 of CCC-C-467. The bias-sewn tubing specified shall conform to MIL-T-40625. Bales shall have the open ends closed with two double-looped wire ties. The wire ties shall be not less than 6 inches long, of 0.072-inch thick galvanized soft iron or steel wire with a 1/2-inch diameter formed eye at each end. The first wire tie shall be applied as close to the bale as possible so there will be a minimum of slack in the tube, the cloth being gathered evenly to form an ear, and all edges of the cloth being 3 to 5 inches from the wire tie at the base of the ear. The second wire tie shall be applied at a distance approximately 1 inch from the first wire tie. The twisted ends of the wire ties shall alternate and face in opposite directions. Each bale shall be reinforced with two flat steel straps conforming to type I, class A, 5/8-inch wide by 0.015-inch thick of QQ-S-781. The straps shall be applied around the bale placed approximately 10 inches in from each end. Sufficient tension shall be applied to the strapping, insuring tight application. Alternatively, each waterproofed roll will be packed in a snug-fitting fiberboard shipping container conforming to class 2, V3c or V3s, style FOL of PPP-B-636. All seams, corners, and manufacturer's joint shall be sealed by the application of 3-inch minimum width olive drab color cloth tape conforming to type III, class 1 of PPP-T-60. Each fiberboard shipping container shall be closed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification. Weight of the contents of each bale shall not exceed 100 pounds with a 10-pound plus

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tolerance permitted, and for fiberboard shipping containers the weight limitations shall be in accordance with the requirements of the container specification.

5.2.2 *Level B.* Each roll of leather, packaged as specified in 5.1, shall be baled in bias-sewn tubing or a bag presewn one end made from cotton osnaburg cloth conforming to type I, class 2, finish A of CCC-C-429 or jute (or kenaf) burlap cloth conforming to class 1 of CCC-C-467. The bias-sewn tubing specified shall conform to MIL-T-40625. Bales shall have the open ends closed with two double-looped wire ties. The wire ties shall be not less than 6 inches long, of 0.072-inch thick galvanized soft iron or steel wire with a 1/2-inch diameter formed eye at each end. The first wire tie shall be applied as close to the bale as possible so there will be a minimum of slack in the tube, the cloth being gathered evenly to form an ear, and all edges of the cloth being 3 to 5 inches from the wire tie at the base of the ear. The second wire tie shall be applied at a distance approximately 1 inch from the first wire tie. The twisted ends of the wire ties shall alternate and face in opposite directions. Each bale shall be reinforced with two flat steel straps conforming to type I, class A, 5/8 inch wide by 0.015 inch thick of QQ-S-781. The straps shall be applied around the bale placed approximately 10 inches from each end. Sufficient tension shall be applied to the strapping, to insure tight application. Alternatively, each roll will be packed in a snug-fitting fiberboard shipping container conforming to type I or II, class 1, style OSC (overlap not less than 3 inches) of PPP-B-636. Each fiberboard shipping container shall be closed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification. Weight of the contents of each bale shall not exceed 100 pounds with a 10-pound plus tolerance permitted and for fiberboard shipping containers the weight limitations shall be in accordance with the requirements of the container specification.

5.2.3 *Level C.* Leather packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** When impregnated with a compound applicable to conditions of intended use, leather conforming to this specification may be used for packing reciprocating and rotating parts of hydraulic machinery. When temperatures exceed 50° Centigrade, vegetable tanned leather should not be specified. It is believed that tannage II, class 1, leather would cause no appreciable corrosion in ordinary installations; tannage II, class 2, leather should be specified only for delicate mechanisms or when operating conditions warrant its use.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number and date of this specification.

(*b*) Tannage required (see 1.2.1).

(*c*) Weight, thickness required (see 3.6.1).

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(*d*) When the fungicide paranitrophenol is required (see 3.7.1).

(*e*) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 In most cases it will be necessary to add waxes, resins or other materials to the leather before it is suitable for use as packing. This may be required by the supplier if so specified in the invitation for bids. Under these conditions the requirements for chloroform extract (see 3.7.2) would not be applicable to the finished packing.

6.4 **Supersession data.** This specification supersedes Federal Specifications: KK-L-157 dated July 31, 1956; KK-L-163 dated January 2, 1957; and KK-L-231 dated November 8, 1949. The comparison of types and tannages is as shown:

KK-L-164

Tannage I—Vegetable Tanned KK-L-157

Tannage II—Chrome tanned:

Class 1—Regular ..... Type I—Regular of KK-L-163

Class 2—Non-corrosive ..... Type II—Non-corrosive of KK-L-163

Tannage III — Chrome — Vegetable-retanned.....KK-L-231

6.5 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Leather, cattlehide, in the original tanning shapes.

Carload minimum weight, 30,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Leather, cattlehide, in the original tannery shapes.

Truckload minimum weight, 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—GL

Navy—S&A

Air Force—ASD

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